Course: Advance Database systems course
Project Title: Online School Management System (OSMS)
Semester: 1st 2026/2027
Total Marks: 20 Points

Project Description (Advanced Database Course):

This assignment aims to apply advanced database design and implementation concepts to a real-world business scenario. Students will design and develop a comprehensive database system that demonstrates effective data modeling, normalization, and integration with a user interface. The project will include the creation of an Entity–Relationship (ER) diagram, logical and physical database design, implementation of advanced SQL features such as views, stored procedures, transactions, and data security considerations.

Students are required to submit a detailed project report that includes a problem statement, project objectives, system scope, ER diagram, database schema, sample queries, screenshots of the implemented system, and testing results. The goal is to provide practical experience in designing, developing, and managing robust, efficient, and secure database-driven applications.

Scenario:

A growing educational academy currently manages student admissions, course registrations, faculty assignments, grade rosters, and tuition fee payments across disparate electronic spreadsheets and manual paper logs. This decentralized approach causes frequent grade discrepancies, double-booking of classroom capacity, unrecorded fee balances, and inefficient administrative tracking.

The academy aims to replace this error-prone system with a centralized **Online School Management System (OSMS)** relational database architecture. The new web or desktop GUI application will empower administrators, instructors, and students to securely manage academic records, process course enrollments, compute GPAs, track tuition transactions, and generate real-time institutional reports.

Objectives:

- **Database Design & Architecture:** Conceptualize and implement an normalized relational schema for academic entities (students, courses, departments, instructors, enrollments, and payments).
- **GUI Development:** Build a GUI application supporting full CRUD operations for administrators, faculty, and students.

- **Automated Academic Rules:** Develop stored procedures and triggers to enforce prerequisite checking, prevent over-enrollment beyond classroom capacity, and update student cumulative GPAs automatically.
- **Reporting Engine:** Generate dynamic analytical reports including student transcript summaries, department enrollment statistics, instructor teaching loads, and unpaid tuition accounts.
- **Transaction Management & Security:** Enforce ACID properties for multi-table transactions (e.g., enrollment drop/add, tuition payments) and implement role-based data access controls.

Your Project report **must be organized and contain the following sections:**

Section	Description
1. Title Page	Title, student names, ID, course, instructor, date
2. Problem Statement	Describe the current school system problems
3. Objectives	General and advanced database objectives
4. System Analysis	Users, requirements, and data flow
5. ER Diagram	Use the given schema to explain entity relationships
6. Database Design	Table structures, constraints, and relationships
7. GUI Design	Screenshots or wireframes of the application
8. Implementation	What tools used to build DB and GUI

Submission Guidelines

- Submit one **PDF document** with all content and figures embedded.
- All diagrams using appropriate tools (ER, Database Schema).
- This is a Team work of (2-4 students)
- Document submission and presentation of work will start after [15/11/2026].